

# DISCLOSURE OF PRE-EXISTING / THIRD-PARTY / OPEN-SOURCE MATERIAL

Submitted in accordance with Section 10 & Clause 1.3 of the Hackathon Student Undertaking

| MATERIAL / COMPONENT                                           | SOURCE / OWNER               | LICENCE / PERMISSION               | PURPOSE                                                                                                                  |
|----------------------------------------------------------------|------------------------------|------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>YOLO11 Pose Model</b><br>yolo11m-pose.pt / yolo11n-pose.pt  | Ultralytics Inc.             | AGPL-3.0 / Open Source             | Multi-person 17-keypoint skeletal pose tracking for candidate posture, head yaw/pitch, and reaching detection.           |
| <b>YOLO11 Object Model</b><br>yolo11m.pt / yolo11n.pt          | Ultralytics Inc.             | AGPL-3.0 / Open Source             | Real-time identification of prohibited items (cell phones, smart devices, chits, and unauthorized materials).            |
| <b>MediaPipe Hand Landmarker</b><br>hand_landmarker.task       | Google LLC                   | Apache License 2.0                 | 21-point 3D hand keypoint extraction to detect phone holding posture and candidate paper-chit pinching gestures.         |
| <b>MediaPipe Gesture Recognizer</b><br>gesture_recognizer.task | Google LLC                   | Apache License 2.0                 | Fine-grained hand gesture classification on candidate crops for anomalous hand activity heuristics.                      |
| <b>OpenCV (opencv-python)</b><br>cv2 v4.8+                     | OpenCV Team                  | Apache License 2.0                 | Video stream ingestion, frame scaling, cropping candidate bounding boxes, and HUD/overlay canvas rendering.              |
| <b>PyTorch (torch)</b><br>torch v2.0+                          | Meta / PyTorch Foundation    | Modified BSD                       | Deep learning runtime backend and accelerated tensor execution across CPU / CUDA / MPS hardware.                         |
| <b>FastAPI</b><br>fastapi v0.100+                              | Tiangolo (Sebastián Ramírez) | MIT License                        | Asynchronous backend REST API framework for configuration, session management, and telemetry endpoints.                  |
| <b>Uvicorn</b><br>uvicorn v0.23+                               | Encode OSS                   | BSD 3-Clause                       | High-performance ASGI server for delivering multipart MJPEG live video proctoring stream to the frontend.                |
| <b>NumPy</b><br>numpy v1.24+                                   | NumPy Developers             | BSD 3-Clause                       | Vector mathematics, Euclidean distance, 3D coordinate geometry, head yaw/pitch angles, and spatial IOU.                  |
| <b>PyYAML &amp; Pydantic</b><br>pyyaml v6.0+ / pydantic v2.0+  | Open Source Community        | MIT License                        | Declarative YAML configuration management (config.yaml) and strict data schema parsing/validation.                       |
| <b>SQLite (sqlite3)</b><br>Python Standard Library             | Public Domain                | Public Domain                      | Local relational database for persistent session storage, detected suspicious events, and audit logs.                    |
| <b>React &amp; ReactDOM</b><br>react.min.js / react-dom.min.js | Meta Platforms, Inc.         | MIT License                        | Declarative single-page proctoring dashboard UI for live monitoring, seat grid status, and anomaly alerts.               |
| <b>Babel Standalone</b><br>babel.min.js                        | Babel Team                   | MIT License                        | Client-side JSX and ES6 script transpilation directly inside the browser runtime without build dependencies.             |
| <b>Hackathon CCTV Datasets</b><br>DRISHTI AI DEXIT Datasets    | DEXIT Global Limited         | Proprietary (Hackathon Authorized) | Official CCTV footage used for pipeline calibration, rule validation, multi-camera testing, and anomaly verification.    |
| <b>AI Coding Assistants</b><br>Claude / Gemini / Antigravity   | Anthropic / Google DeepMind  | Permitted under Clause 6           | Assistance in algorithmic scaffolding, automated unit testing harness formulation, and technical documentation drafting. |